

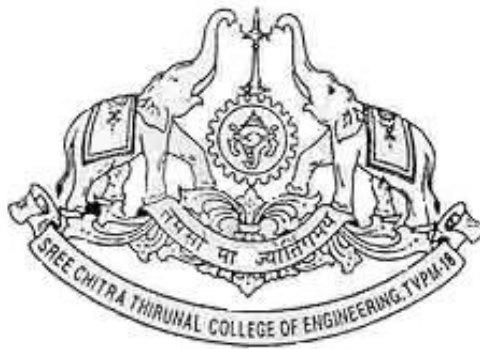
A PROJECT REPORT ON
WEAR ANALYSIS OF DISC BRAKES

*Submitted in partial fulfilment of the requirements for the
award of the degree of*

BACHELOR OF TECHNOLOGY

In

MECHANICAL (AUTOMOBILE) ENGINEERING



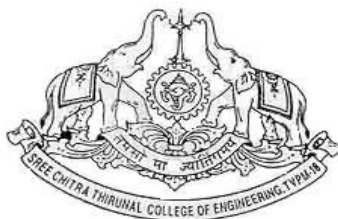
Submitted by

MANU P NAIR	12 402 038
R ANANTHAKRISHNAN	12 402 045
SIDHARTH S	12 402 055
VAISHAKH G	12 402 059

DEPARTMENT OF MECHANICAL ENGINEERING
SREE CHITRA THIRUNAL COLLEGE OF ENGINEERING
THIRUVANANTHAPURAM 695 018

MAY 2016

**SREE CHITRA THIRUNAL COLLEGE OF ENGINEERING,
THIRUVANANTHAPURAM – 695 018.
DEPARTMENT OF MECHANICAL ENGINEERING.**



CERTIFICATE

Certified that project work entitled “**WEAR ANALYSIS OF DISC BRAKES**” is a bonafide work carried out in the eighth semester by “**MANU P NAIR (12402038), R ANANTHAKRISHNAN (12402045), SIDHARTH S (12402055), VAISHAKH G (12402059)**” in partial fulfilment for the award of Bachelor of Technology in “**MECHANICAL (AUTOMOBILE) ENGINEERING**” from University of Kerala during the academic year 2015-2016 who carried out the project work under the guidance and no part of this work has been submitted earlier for the award of any degree.

PROJECT CO-ORDINATORS

E.JAYAKUMAR

Assistant Professor
Department of Mechanical Engg.
SCT College of Engineering
Thiruvananthapuram - 18.

INTERNAL GUIDE

T S SIGIKUMAR

Assistant Professor
Department of Mechanical Engg.
SCT College of Engineering.
Thiruvananthapuram – 18.

HEAD OF THE DEPARTMENT

Dr. AJITH R

Department of Mechanical Engg.
SCT College of Engineering,
Thiruvananthapuram – 18

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ABSTRACT

In the technologically advanced era that we currently live in, there is a growing demand for cheaper and durable materials for a variety of purpose. The most important safety feature of an automobile is its braking system. The ability of a braking system is to provide a safe and repeatable stopping, which is related to safety of automobiles in humans. Automotive friction materials are multi component composites in which binder, fibre, fillers and property modifiers play an important role.

In this paper we will be doing an investigative study on the non-uniform wear rate (wear depth) of a few brake pad materials and to analyze the wear in terms of surface wear depth occurring in each of them and to determine which is more wear resistant during various operating conditions and atmospheric condition. For this purpose, the materials we have chosen for the pad wear Alumina (A479M), Carbon fibre composite, Epoxy Resin polyester composite and for the disc the material we have chosen is AMC. Furthermore, we will also be comparing these results which we obtained for the different brake pad materials and find which one of them is more wear resistant for a chosen driving condition.

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CHAPTER 1

INTRODUCTION

A disc brake is a type of brake that uses calipers to squeeze pairs of pads against a disc in order to create friction that retards the rotation of a shaft, such as a vehicle axle, either to reduce its rotational speed or to hold it stationary. The energy of motion is converted into waste heat which must be dispersed. Hydraulic disc brakes are the most commonly used form of brake for motor vehicles but the principles of a disc brake are applicable to almost any rotating shaft. Disc brake systems operate by turning the kinetic energy of a car in motion into heat. Brake pads are pressed against iron rotors attached to the car's wheels. There is friction between the pads and the rotors, and this friction causes the car to slow as its momentum is dissipated as heat in the pads and rotors. Over time, this friction causes brake pads to wear. When enough of them has worn away, they need to be replaced.

Wear is the process of the gradual removal of material from solid surfaces that are subjected to sliding contact. It is a complex phenomenon that is relevant to many problems involving frictional contact, such as mechanical brakes, seals, metal forming, and orthopedic implants. The rate of wear depends on the properties of the contacting materials and operating conditions.

Why would pads wear unevenly?

The areas of a brake pad called upon to do more work, or which run hotter, will wear away faster. Although pressing a pad against a rotor may seem to distribute the load fairly evenly, it does not.

Uneven wear occurs in roughly three different forms:

- longitudinal taper,
- radial taper, and
- inner vs. outer pad depth.

The leading edge of a brake pad is the one that may be thought of as first touching the rotor as it turns. If your calipers are mounted behind the wheel axis, the bottom pad edges will lead. The friction of the pad against the turning rotor causes the pad itself to try to rotate about an imaginary axis behind it, with the effect that the leading edge of the pad presses

harder against the rotor, doing more work and getting hotter. This uneven heating leads to longitudinal wear: uneven pad depth from leading to trailing edges.

Radial taper wear can be seen as a difference in pad depth along the radius of the rotor. A brake caliper is mounted to its strut on one edge. The forces created by applying the brake pedal cause yet another twisting moment, this time of the caliper assembly around an axis through its mounting points. There are other factors as well: a square-ish pad against a round rotor means that the radially inner part of the rotor will spend proportionally more time under the pad, and less in cooling air. Heat differences also exist because the inner edge of the rotor interfaces with the hub rather than air, limiting conduction.

1.1 About the Software- COMSOL MULTIPHYSICS 5.0

COMSOL Multiphysics is a finite element analysis, solver and Simulation software / FEA Software package for various physics and engineering applications, especially coupled phenomena, or multiphysics. Comsol is an interactive program for solving coupled Partial Differential Equation (PDE) in one or more physical domains simultaneously (e.g., electrostatics and heat transfer). Comsol has strong links to Matlab which is handy for for such things as coding functions in your models, for automating tasks, and for custom pre- and post-processing, etc. Comsol can be used either through a Graphical User Interface (GUI) or by calling Comsol functions from a Matlab script, or both.

Add-on modules are available. These provide predefined models and functions for specific applications to save effort in programming the base product. Ten add-ons have been licensed at the University of Waterloo:

- Chemical Engineering Module
- Reaction Engineering Lab (for Comsol Script)
- Earth Sciences
- Alternating Current (AC)/Direct Current (DC) Module
- Micro Electro-Mechanical Systems (MEMS) Module
- Structural Mechanics Module
- Heat Transfer Module
- Radio Frequency (RF) Module
- Computer Aided Design (CAD) Import Module

- Comsol Script

Using comsol multiphysics, we used a wear model to predict the wear in an automotive disc brake. The figure shown below shows a Comsol Multiphysics 5.0 console window:

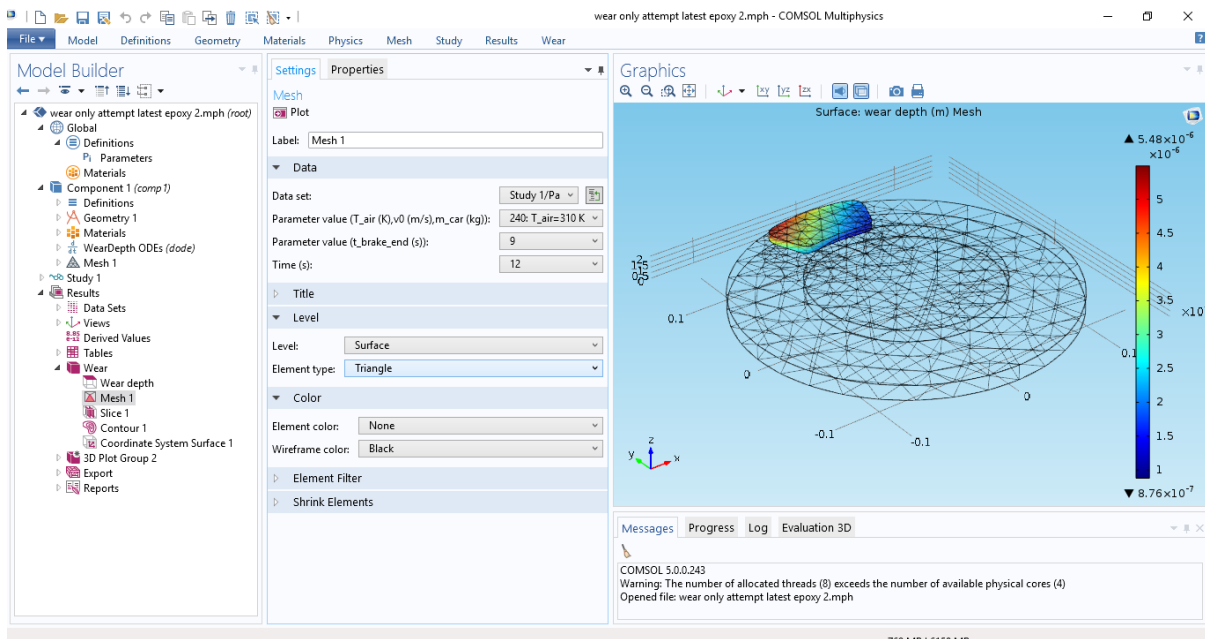


Fig 1: Comsol Multiphysics 5.0. console

CHAPTER 2

LITERATURE SURVEY

1. **N. H. Elabbasi, M. J. Hancock, S. B. Brown** illustrated a contact offset based wear model implementation by simulating wear in automotive disc brakes. They developed a 3D thermal structural disc brake model involving representative brake disc/rotor and brake pads. These problems are coupled through a frictional heat generation, thermal expansion and thermal contact. They have implemented Archard's wear law as a boundary ordinary differential equation in Comsol Multiphysics with the wear depth/thickness as the independent variable.
2. **Hiroshi Yoshida, Kenichi Yamazaki, Morio Yumoto** speculated that the disc brake for vehicles is characterized in that either the disc adapted to rotate integrally with the wheels or the brake pads which are brought into friction contact with each face of the disc are made of a carbon reinforced ceramic composite. They conducted two experiments with carbon fiber reinforced ceramic composite being used as brake pads and discs in either cases.

CHAPTER 3

OBJECTIVES AND IMPLEMENTATION

3.1 OBJECTIVES

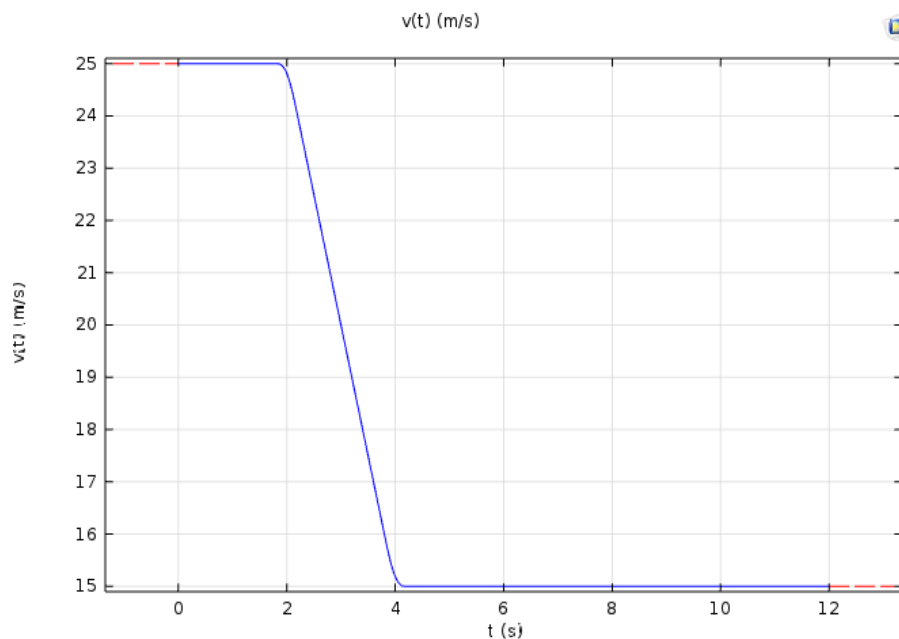
The major objectives of this project are given below:

To run a simulation showing the non-uniform wear rate on brake pads in an automobile and to obtain the following:

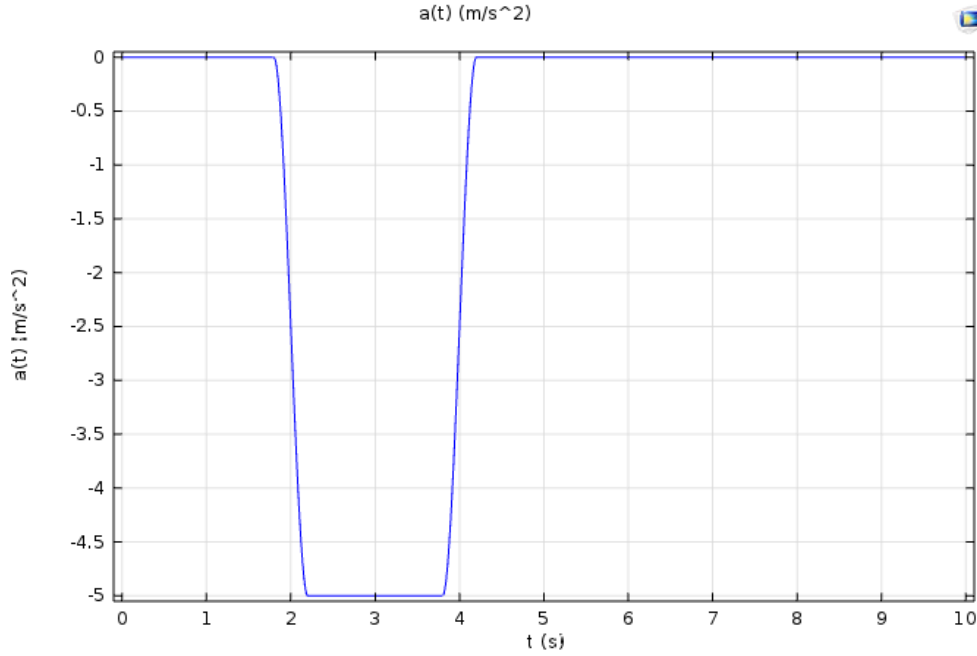
- wear rate of different brake pad materials.
- wear rate of these pads during different operating and climatic conditions

To conduct an investigative study of the wear rate of the different materials during these conditions described above and learn which among them is more wear resistant.

A 2200 kg car is travelling at 25 m/s (90 kmph or about 65 miles per hour), until the driver suddenly panic brakes for 2 secs. AT that point the eight brake pads slows the car down at rate of 5 m/s^2 (assuming the wheels do not skid against the road). Upon braking for 2 seconds the driver releases the brakes, leaving the car travelling at 15 m/s for 8 secs without engaging the brakes.



Graph 3.1: Velocity vs time graph showing the change in velocity during panic braking



Graph 3.2: Acceleration vs time graph showing the change in acceleration during panic braking

3.2 IMPLEMENTATION

Archard's law is a simple but widely used wear law that relates the volume of material removed due to wear W to the normal contact force F_N , sliding distance L_T , material hardness H , and a material-related constant K .

$$W = \frac{KF_N}{H} L_T$$

We modified this law toward wear depth as:

$$H(T) \frac{dw}{dt} = k P_N V_S$$

$H(T)$: hardness depending on the material temperature [Pa],

w : wear depth [m],

P_N : normal contact pressure vector [Pa],

V_S : sliding velocity vector [m/s]

k : modify constant compared to practical wear data (material constant)

In the source term $P_N V_S$ is scalar multiplication of two vectors.

Then representing the vector as Cartesian components, we get:

$$P_N = \{P_x, P_y, P_z\}$$

$$V_S = \{V_x, V_y, V_z\}$$

It is also worth mentioning the fact that the component has a x-direction and is the pressure component applied to the yz-surface. Now we will represent the ODE $H(T) \frac{dw}{dt} = kP_N V_S$ to the term of Cartesian Components.

$$H(T) \frac{dw}{dt} = P_x V_x + P_x V_y + P_x V_z + P_y V_x + P_y V_y + P_y V_z + P_z V_x + P_z V_y + P_z V_z$$

At the point of practical view, 3 component of the pressure must be calculated from structure-mechanic and thermal stress in the wear space. For simplifying the problem, we considered this object as a macro phenomenon, so the pressure function is a component of coordinates and other factors without theoretical contradiction. By deriving for the velocity component, we get:

$$V_x = \frac{xv(t)}{R}$$

$$V_y = \frac{-yv(t)}{R}$$

$$V_z = 0$$

Here, the parameter R is the car's wheel radius and v(t) is the car's speed when it brakes. We skipped the derivation process about the thermal pressure from kinetic energy and we used the coordinate components of the pressure and got the following:

$$P_x = \frac{m_{car} \frac{dV_x(t)}{dt}}{8\mu}$$

$$P_y = \frac{m_{car} \frac{dV_y(t)}{dt}}{8\mu}$$

$$P_z = \frac{-m_{car} G(z) a(t)}{8\mu}$$

Here, m_{car} is the car's mass, μ is the friction rate (it was setting 0.3) for process between pad's top surface an automotive rotor's brake hole, $G(z)$ is the weight function for z-direction pressure.

Next, the destination of the weight function $G(z)$ is the experimental correction for the penetration between the contact surface. We set the function form of $G(z)$ from various reference media and based on the assumption that the attenuation function shapes of the contact pressure is exponential in depth.

Function $G(z)$ is the following:

$$G(z) = \exp[-100 \times (z_{down} - z)]$$

Here, z_{down} is the height (z value) of the pad bottom, rate 100 is unit conversion rate (m->Cm). The spatial wear resulting is accuracy at this function that fits how well for the surface penetration of pressure (contact pressure) and it depends on material. Also here, the pressure components mean that the pad is incompressible media and isotropic in this assumption.

Currently we are researching about the x and y component of the pressure. In this geometrical model, P_x and P_y factors must be considered in the 4 surfaces except pad's top surface in the pad's domain.

In COMSOL 5.0 there are 3 series of hardness, Microhardness, Vickers hardness and Brinell hardness. These are needed for the unit measures conversion.

In non-thermal case, hardness is setting 800[MPa] for the pad as microhardness. So the hardness depends on the temperature that this function form is setting following from the preceding practical research works.

$$Hd = H_{initial} - T^2$$

Here, is 800[MPa] as the non-thermal hardness, T[K] is the pad's temperature.

CHAPTER 4

MATERIAL PROPERTIES & PARAMETERS

4.1 MATERIAL PROPERTIES

The pad we used for running the first simulation was Alumina a479m [6] fine ceramic and its material properties were:

- Thermal conductivity – 32 W/m.K
- Specific heat – 780 J/kg.K
- Density – 3900 kg/m³
- Hardness – 154 Mpa (15.7 HV units)

The material properties of the disc (AMC reinforced with SiC) were the following:

- Thermal Conductivity – 82 W/m.K
- Specific heat – 449 J/kg.K
- Density – 7870 kg/m³

The brake pad material we have chosen for running the initial simulation is fine ceramic alumina (A479M).

Apart from the materials used earlier, we also used the following materials for the brake pad:

Carbon fibre composite

- Thermal conductivity: 117.6 W/mK
- Specific heat: 875.0412 J/kgK
- Density: 2767.99 kg/m³
- Hardness: 1.34 GPa

Epoxy resin polyester composite

- Thermal conductivity: 1.66 W/mK
- Specific heat: 1000 J/kgK

- Density: 6000 kg/m³
- Hardness: 0.0161 Gpa

4.2 PARAMETERS

NAME	EXPRESSION	VALUE	DESCRIPTION
mu	0.35	0.35	Friction Coefficient
v0	25[m/s]	25 m/s	Initial Vehicle Speed
a0	-5[m/s ²]	-5 m/s ²	Initial Vehicle acc
r_wheel	0.25[m]	0.25	Wheel radius
m_car	2200[Kg]	2200	Vehicle mass
t_brake_start	2[s]	2	Braking time start
t_brake_end	4[s]	4	Braking time end
T_air	300[K]	300	Temperature air
t_calc_end	12[s]	12	Calculation time
T_end	1200[K]	1200	Temperature (end)
H_initial	0.154e9[Pa]	0.154e9	Initial hardness
mu_pad	0.35	0.35	Friction rate for pad
K_pad	1	1	Modify rate for pad

Table 4: Parameters for Alumina a479m

The above tables show the parameter used when Alumina a479m was used as the brake pad material.

In order to simulate for carbon fibre composite and epoxy brake pad materials we used the same parameters as that of alumina a479m with only exemption being the initial hardness value and temperature value. The values were 1.3400E9 [Pa] for initial hardness and 600 [K] for temperature(end) for the case of carbon fibre composite respectively. Similarly in the case of epoxy resin polyester composite initial hardness is taken as 1.6100E7 [Pa] and the temperature(end) is taken as 500 [K]. Also mu values were changed as 0.23 and 0.29 respectively for carbon fibre composite and epoxy resin polyester composite.

CHAPTER 5

SIMULATION MODELLING

5.1 DESIGN PROCEDURE

From the File menu, choose New.

5.1.1 MODEL WIZARD

1. In the New window, click Model Wizard. In the Model Wizard window, click 3D.
2. In the Select physics tree, select Heat Transfer>Heat Transfer in Solids (ht).
3. Click Add.
4. Click Study.
5. In the Select study tree, select Preset Studies>Time Dependent.
6. Click Done.

Parameters

1. On the Home toolbar, click Parameters.
2. In the Settings window for Parameters, locate the Parameters section.
3. Type in the parameter which has been mentioned above.

5.1.2 GEOMETRY 1

Cylinder 1 (cyl1)

1. On the Geometry toolbar, click Cylinder.
2. In the Settings window for Cylinder, locate the Size and Shape section.
3. In the Radius text field, type 0.14.
4. In the Height text field, type 0.013.

Cylinder 2 (cyl2)

1. Right-click Cylinder 1 (cyl1) and choose Build Selected.
2. On the Geometry toolbar, click Cylinder.

3. In the Settings window for Cylinder, locate the Size and Shape section.
4. In the Radius text field, type 0.08.
5. In the Height text field, type 0.01.
6. Locate the Position section. In the z text field, type 0.013.

Work Plane 1 (wp1)

1. Right-click Cylinder 2 (cyl2) and choose Build Selected.
2. On the Geometry toolbar, click Work Plane.
3. In the Settings window for Work Plane, locate the Plane Definition section.
4. In the z-coordinate text field, type 0.013.
5. Right-click Work Plane 1 (wp1) and choose Show Work Plane.

Bézier Polygon 1 (b1)

1. On the Geometry toolbar, click Primitives and choose Bézier Polygon.
2. In the Settings window for Bézier Polygon, locate the Polygon Segments section.
3. Find the Added segments subsection. Click Add Cubic.
4. Find the Control points subsection. In row 1, set yw to 0.135.
5. In row 2, set xw to 0.02.
6. In row 2, set yw to 0.135.
7. In row 3, set xw to 0.05.
8. In row 3, set yw to 0.13.
9. In row 4, set xw to 0.04.
10. In row 4, set yw to 0.105.
11. Find the Added segments subsection. Click Add Cubic.
12. Find the Control points subsection. In row 2, set xw to 0.03.
13. In row 2, set yw to 0.08.
14. In row 3, set xw to 0.035.

15. In row 3, set yw to 0.09.
16. In row 4, set xw to 0.
17. In row 4, set yw to 0.09.
18. Find the Added segments subsection. Click Add Cubic.
19. Find the Control points subsection. In row 2, set xw to -0.035.
20. In row 3, set xw to -0.03.
21. In row 3, set yw to 0.08.
22. In row 4, set xw to -0.04.
23. In row 4, set yw to 0.105.
24. Find the Added segments subsection. Click Add Cubic.
25. Find the Control points subsection. In row 2, set xw to -0.05.
26. In row 2, set yw to 0.13.
27. In row 3, set xw to -0.02.
28. In row 3, set yw to 0.135.
29. Click Close Curve. To complete the pad cross section, you must make the top-left and top-right corners sharper. Do so by changing the weights of the Bézier curves.
30. Find the Added segments subsection. In the Added segments list, select Segment 1 (cubic).
31. Find the Weights subsection. In the 3 text field, type 2.5.
32. Find the Added segments subsection. In the Added segments list, select Segment 4 (cubic).
33. Find the Weights subsection. In the 2 text field, type 2.5.
34. Right-click Bézier Polygon 1 (b1) and choose Build Selected.

Extrude 1 (ext1)

1. On the Geometry toolbar, click Extrude.
2. In the Settings window for Extrude, locate the Distances from Plane section.
3. In the table, enter the following settings: Distances - 0.0065m
4. Right-click Extrude 1 (ext1) and choose Build Selected. The model geometry is now complete. Next, define some selections of certain boundaries. You will use them when defining the settings for component couplings, boundary conditions, and so on.

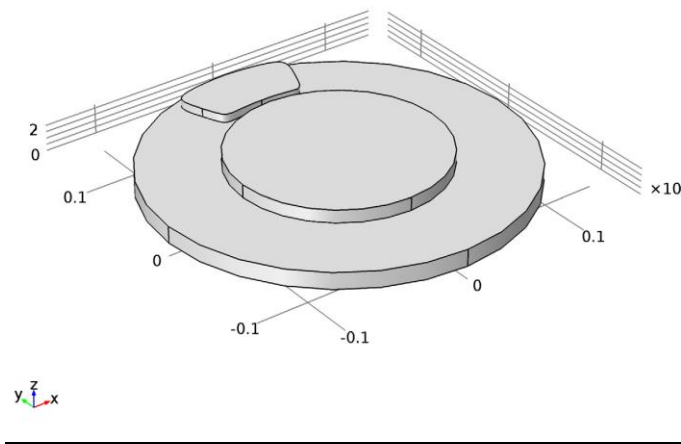


Fig 5: Disc brake model

5.1.3 DEFINITIONS

Explicit 1

1. On the Definitions toolbar, click Explicit.
2. In the Settings window for Explicit, type Disc faces in the Label text field.
3. Locate the Input Entities section. From the Geometric entity level list, choose Boundary.
4. Select Boundaries 1, 2, 4–6, 8, 13–15, and 18 only.

Explicit 2

1. On the Definitions toolbar, click Explicit.
2. In the Settings window for Explicit, type Pad faces in the Label text field.

3. Locate the Input Entities section. From the Geometric entity level list, choose Boundary.
4. Select Boundaries 9, 10, 12, 16, and 17 only.

Explicit 3

1. On the Definitions toolbar, click Explicit.
2. In the Settings window for Explicit, type Contact faces in the Label text field.
3. Locate the Input Entities section. From the Geometric entity level list, choose Boundary.
To select the contact surface boundary, it is convenient to temporarily switch to wireframe rendering.
4. Click the Wireframe Rendering button on the Graphics toolbar.
5. Select Boundary 11 only.
6. Click the Wireframe Rendering button on the Graphics toolbar.

Explicit 4

1. On the Definitions toolbar, click Explicit.
2. In the Settings window for Explicit, type External surfaces in the Label text field.
3. Locate the Input Entities section. Select the All domains check box.
4. Locate the Output Entities section. From the Output entities list, choose Adjacent boundaries. These instructions make you select the external boundaries of the wheel and the pad.

Integration 1 (intop1)

1. On the Definitions toolbar, click Component Couplings and choose Integration.
2. In the Settings window for Integration, locate the Source Selection section.
3. From the Geometric entity level list, choose Boundary.
4. From the Selection list, choose Contact faces.
5. Select Boundary 11 only.

Integration 2 (intop2)

1. On the Definitions toolbar, click Component Couplings and choose Integration.
2. In the Settings window for Integration, locate the Source Selection section.
3. From the Geometric entity level list, choose Boundary.
4. From the Selection list, choose External surfaces.
5. After this step, Define the velocity and acceleration of car through these two piecewise and analytic functions.

Piecewise 1 (pw1)

1. On the Definitions toolbar, click Piecewise.
2. In the Settings window for Piecewise, type v in the Function name text field.
3. Locate the Definition section. In the Argument text field, type t.
4. From the Smoothing list, choose Continuous second derivative.
5. From the Transition zone list, choose Absolute size.
6. In the Size of transition zone text field, type 0.2.
7. Find the Intervals subsection. In the table, enter the following settings:

Start	End	Function
0	t_brake_start	v0
t_brake_start	t_brake_end	$v0 + a0 * (t - t_brake_start)$
t_brake_end	12	$v0 + a0 * (t_brake_end - t_brake_start)$

Table 5.1: Velocity function table

8. Locate the Units section. In the Arguments text field, type s.
9. In the Function text field, type m/s.
10. Click the Plot button.

Analytic 1 (an1)

1. On the Definitions toolbar, click Analytic.

2. In the Settings window for Analytic, type a in the Function name text field.
3. Locate the Definition section. In the Expression text field, type $d(v(t),t)$.
4. In the Arguments text field, type t.
5. Locate the Units section. In the Arguments text field, type s.
6. In the Function text field, type m/s^2 .
7. Locate the Plot Parameters section. In the table, enter the following settings:

Argument	Lower limit	Upper limit
t	0	10

Table 5.2: Time limit table

8. Click the Plot button

P_x (P_x)

1. On the Definitions toolbar, click Analytic.
2. In the Settings window for Analytic, type P_x in the Function name text field and type P_x in the Label text field.
3. Label text field.
4. Locate the Definition section. In the Expression text field, type $d(V_x(t,x),t)*m_{car}/(8*m_{u_pad})$.
5. In the Arguments text field, type t, x.
6. Locate the Units section. In the Arguments text field, type K.
7. Locate the Plot Parameters section. In the Arguments text field, type 0, x_down (-0.05m) in the Lower limit row and type t_calc_end (12s), x_upper (0.05m) in the Upper limit row, respectively.

P_y (P_y)

1. In the Definitions toolbar, click Analytic.
2. In the Settings window for Analytic, type Py in the Function name text field and type P_y
3. Label text field.
4. Locate the Definition section. In the Expression text field, type $d(Vy(t,y),t)*m_car/(8*m_u_pad)$.
5. In the Arguments text field, type t, y.
6. Locate the Units section. In the Arguments text field, type K.
7. Locate the Plot Parameters section. In the Arguments text field, type 0, y_down (0.105m) in the Lower limit row and type t_calc_end (12s) and y_upper (0.135m) in the Upper limit row, respectively.

P_z (Pz)

1. On the Definitions toolbar, click Analytic.
2. In the Settings window for Analytic, type Pz in the Function name text field and type P_z label text field.
3. Locate the Definition section. In the Expression text field, type $-a(t)*Gz(z)*m_car/(8*\mu_pad)$.
4. In the Arguments text field, type t, z.
5. Locate the Units section. In the Arguments text field, type K.
6. Locate the Plot Parameters section. In the Arguments text field, type 0, z_down (0.013m) in the Lower limit row and type t_calc_end (12s), z_upper (0.023m) in the Upper limit row, respectively.

V_x (Vx)

1. On the Defintions toolbar, click Analytic.

2. In the settings window for Analytic, type V_x in the function name text field and type V_x in the label text field.
3. Locate the Definition section. In the expression text field, type $x*v(t)/r_{wheel}$.
4. In the Arguments section, type t, x .
5. Locate the Units section. In the Arguments text field, type s, m .
6. Locate the Plot Parameters section. In the Arguments text field, type $0, x_{down} (-0.05m)$ in the Lower limit row and type $t_{calc_end} (12s), x_{upper} (0.05m)$ in the Upper limit row, respectively.

$V_y(V_y)$

1. On the Definitions toolbar, click Analytic.
2. In the settings window for Analytic, type V_y in the function name text field and type V_y in the label text field.
3. Locate the Definition section. In the expression text field, type $-y*v(t)/r_{wheel}$.
4. In the Arguments section, type t, y .
5. Locate the Units section. In the Arguments text field, type s, m .
6. Locate the Plot Parameters section. In the Arguments text field, type $0, y_{down} (0.105m)$ in the Lower limit row and type $t_{calc_end} (12s), y_{upper} (0.135m)$ in the Upper limit row, respectively.

$G_z(G_z)$

1. On the Definitions toolbar, click Analytic.
2. In the settings window for Analytic, type G_z in the function name text field and type G_z in the label text field.
3. Locate the Definition section. In the expression text field, type $\exp(-100*(0.013-z))$.
4. In the Arguments section, type z .

5. Locate the Units section. In the Arguments text field, type m.
6. Locate the Plot Parameters section. In the Arguments text field, type z_down (0.013m) in the Lower limit row and type z_upper (0.023m) in the Upper limit row, respectively.

Hardness (Hd)

1. On the Definitions toolbar, click Analytic.
2. In the Settings window for Analytic, type Hd in the Function name text field and type Hardness Label text field.
3. Locate the Definition section. In the Expression text field, type $H_initial - (T^2)$.
4. In the Arguments text field, type T.
5. Locate the Units section. In the Arguments text field, type K.
6. Locate the Plot Parameters section. In the Arguments text field, type T_air in the Lower limit row and type T_end in the Upper limit row.

5.1.4 Wear depth ODEs (dode)

First, in Physics toolbar click Add Physics and then click Mathematics->ODE and DAE Interfaces->Domain ODEs and DAEs (dode). In the Settings window, type WearDepth ODEs Label text field. Locate the Domain section. In the Selectn , select manual and choose 3 only.

Distribution ODE 1

1. In the Source Term selection, type $-K_pad * srcTerm(t,x,y,z)$ for f .
2. In the Damping or Mass Coefficient selection, type $Hd(T)$ for da .

5.1.5 MESH 1

Free Triangular 1

1. In the Model Builder window, under Component 1 (comp1) right-click Mesh 1 and choose More Operations>Free Triangular.
2. Click the Transparency button on the Graphics toolbar.
3. Select Boundaries 4, 7, and 11 only.

4. Click the Transparency button on the Graphics toolbar again to return to the original state.

Size

1. In the Model Builder window, under Component 1 (comp1)>Mesh 1 click Size.
2. In the Settings window for Size, locate the Element Size section.
3. From the Predefined list, choose Extra fine.

Swept 1

In the Model Builder window, right-click Mesh 1 and choose Swept.

Distribution 1

1. In the Model Builder window, under Component 1 (comp1)>Mesh 1 right-click Swept 1 and choose Distribution.
2. In the Settings window for Distribution, locate the Distribution section.
3. In the Number of elements text field, type 2.
4. In the Model Builder window, right-click Mesh 1 and choose Build All.

5.1.6 STUDY 1

Step 1: Time Dependent

1. In the Model Builder window, expand the Study 1 node, then click Step 1: Time Dependent
2. In the Settings window for Time Dependent, locate the Study Settings section.
3. In the Times text field, type range (0,0.5,1.5) range (1.55,0.05,3) range (3.2,0.2,5) range (6,1,12).

Solution 1

1. On the Study toolbar, click Show Default Solver.
2. In the Model Builder window, expand the Study 1>Solver Configurations node.

3. In the Model Builder window, expand the Solution 1 node, then click Time-Dependent Solver 1.
4. In the Settings window for Time-Dependent Solver, click to expand the Absolute tolerance section.
5. Locate the Absolute Tolerance section. In the Tolerance text field, type $1e-4$.
6. Click to expand the Time stepping section. Locate the Time Stepping section. From the Steps taken by solver list, choose Intermediate. This setting forces the solver to take at least one step in each specified interval.
7. On the Study toolbar, click Compute.

5.1.7 RESULTS

Wear (ht)

1. In the Results window, click 3D Plot Group .
2. In the Settings window for 3D Plot Group, type Wear the Label text field.
3. Right-click Wear (ht) and click Surface.
4. In the Settings window for Surface, type Wear depth the Label text field.
5. In the Expression selection, type w the Expression text field.
6. In the Expression selection, type m the Unit text field.

CHAPTER 6

RESULTS

6.1 Alumina A479M

Case 1:

The following parameters were taken into consideration, Air temperature is taken to be 310 K, Initial velocity 30 m/s, mass of the vehicle 2400 kg and the braking time as 6 sec. After running the simulation, the following image was obtained showcasing the wear depth in the brake pad.

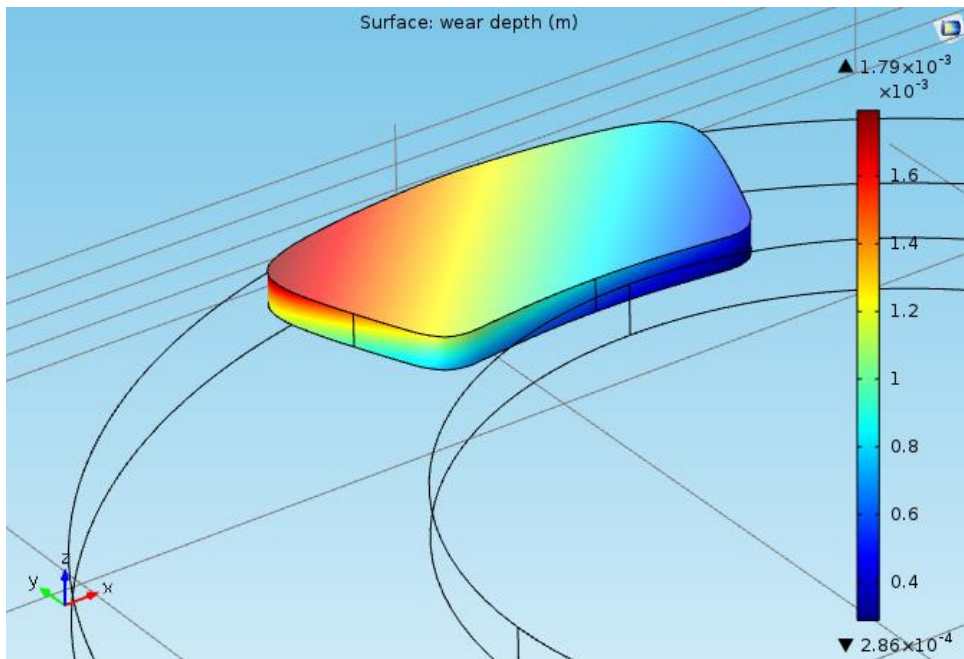


Fig 6.1: Alumina A479M at a temperature of 310K, mass of vehicle 2200kg, initial velocity 25m/s braked for 2s shown 8s after braking

The leading edge has more wear depth than the trailing edge when the disc is rotated in the clockwise direction. The trailing edge is represented as blue region and leading edge as red region.

Case 2:

When the braking time of the vehicle is increased from 2 seconds to 4 seconds, the corresponding wear depth is shown in the figure:

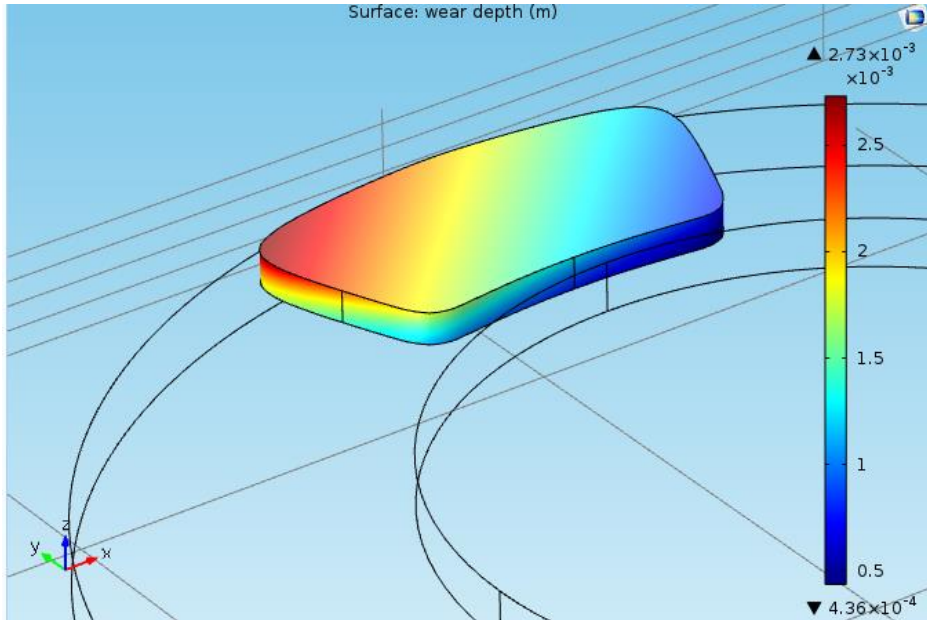


Fig 6.2: Wear depth when the braking time is increased to 4 seconds

Comparing with the earlier figure, it is clear that when the brake is applied for a longer duration the brake pad wears more and hence gives way to comparatively increased wear rate.

Case 3:

In order to understand the impact of velocity on the brake pads we reduced the velocity by half (from 25 m/s to 15 m/s) while keeping all the other parameters same as the initially considered parameter, the corresponding wear depth is obtained.

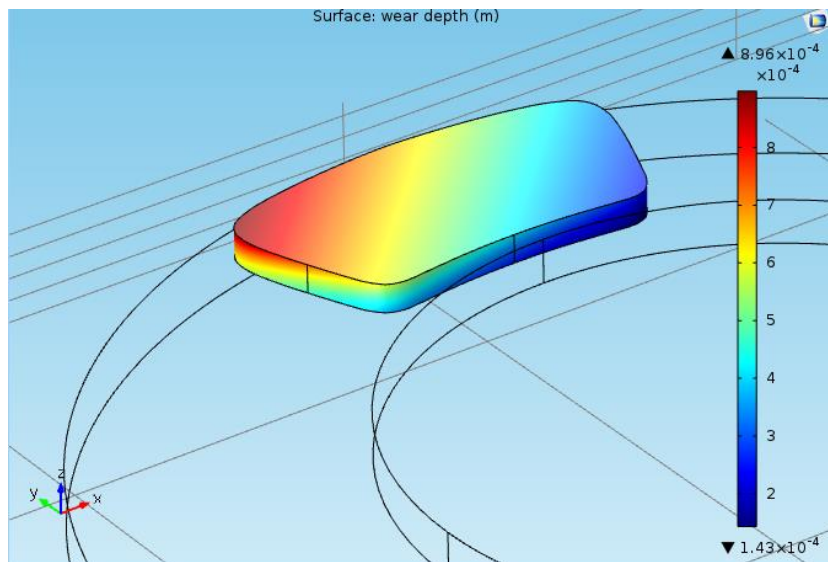


Fig 6.3: Alumina A479M when only the initial velocity is reduced to 15m/s

From the results, we were able to understand the impact of initial velocity on the non-uniform wear rate. We can see in the image that the uneven wear imparted on the pad is drastically reduced as a result of the reduction in velocity.

Case 4:

To understand the effect of vehicles mass on the wear depth (loaded and unloaded condition), vehicles mass is reduced to 1600 kg from 2200 kg while keeping the initially considered parameters as the same (velocity 25 m/s, Temperature 310 K)

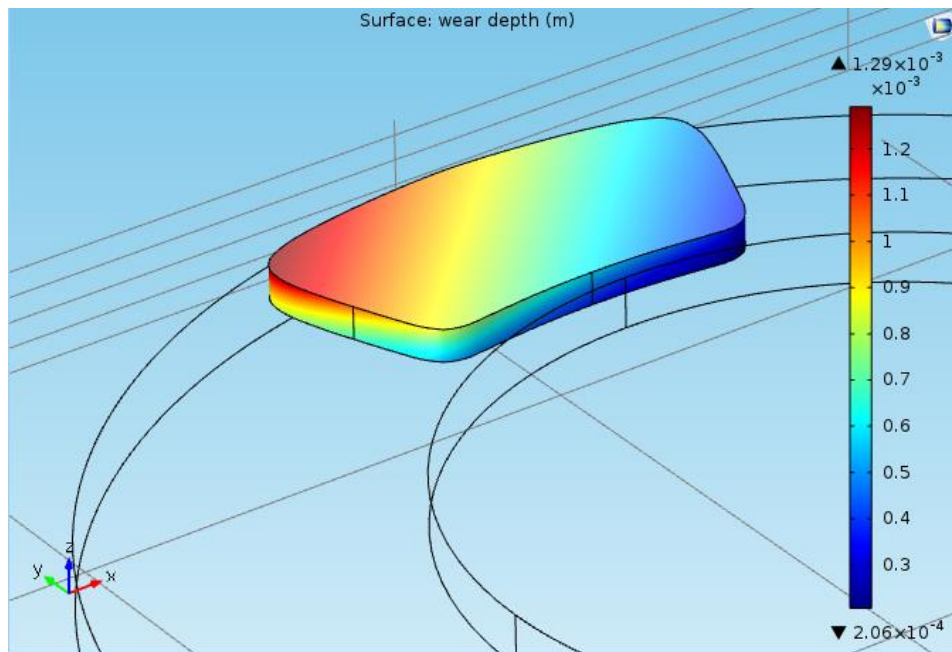


Fig 6.4: Alumina A479M when only the mass is reduced from 2200kg to 1600kg

From the result above, we were able to understand the impact of mass of the vehicle on the non-uniform wear rate. From the image, we can see that though there is a reduction in the non-uniform wear rate when the mass of vehicle is reduced, we can also see that the reduction is not as drastic as the one where the initial velocity is reduced.

Case 5:

In the next case, the surrounding air temperature was reduced from 310K to 275K. This is to learn the impact of the surrounding air temperature for this particular material.

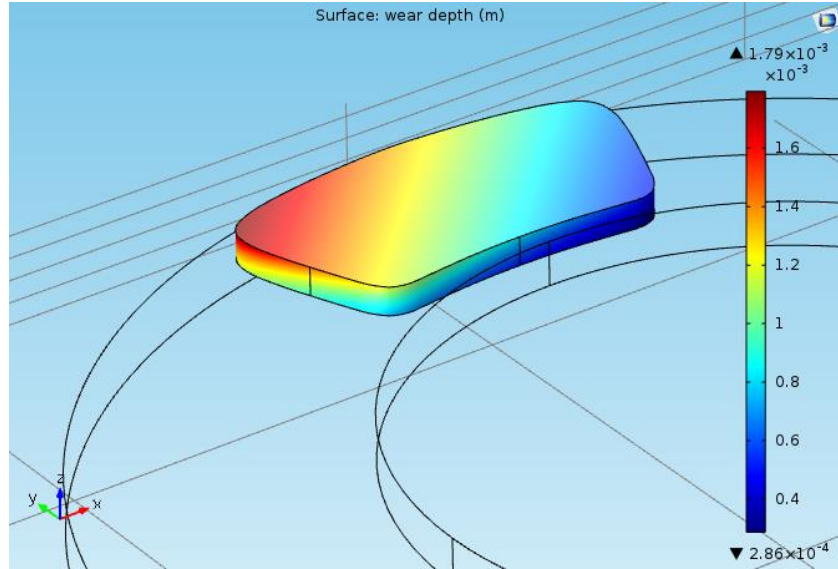


Fig 6.5: Alumina A479M when the air temperature is reduced from 310K to 275K

The result shown above clearly indicates that the surrounding air temperature does not have any major say in the non-uniform wear rate of brake pads. The image exactly resembles the initial Fig 5.1 and so we can conclude that the surrounding air temperature has no impact on the non-uniform wear rate.

Case 6:

The image below shows the wear rate when the air temperature is 275K, mass of the vehicle is 1600kg and the initial velocity of the vehicle is 5m/s. The image also makes it apparent that the wear rate is extremely low such that it can be assumed that no significant wear takes place.

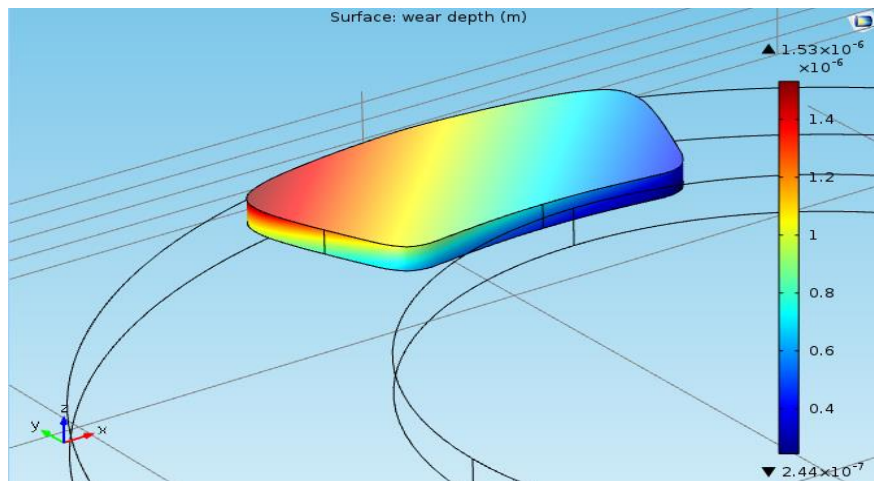


Fig 6.6: Alumina A479M when the air temperature is 275K, mass of the vehicle is 1600kg and the initial velocity of the vehicle is 5m/s (condition where the wear rate is the lowest)

6.2 Carbon fibre Composite

Case 1:

The following parameters were taken into consideration, Air temperature is taken to be 310 K, Initial velocity 25 m/s, mass of the vehicle 2200 kg and the braking time as 2 sec. After running the simulation, the following image was obtained showcasing the wear depth in the brake pad.

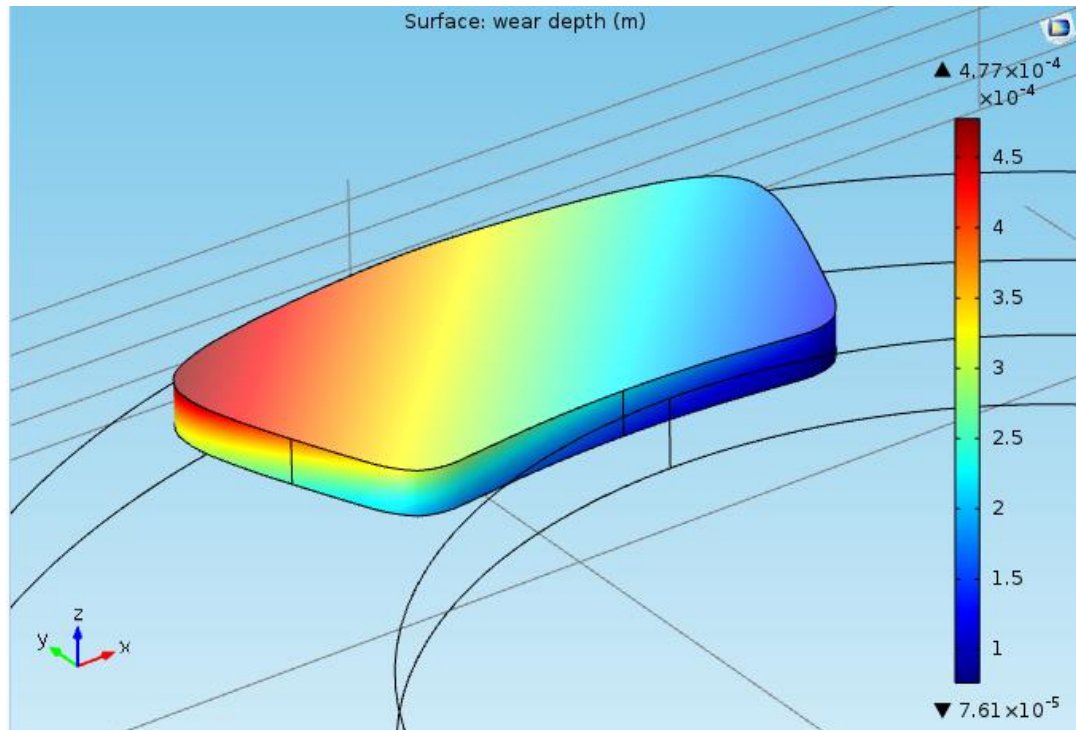


Fig 6.7: Carbon Fibre Composite at a temperature of 310K, mass of vehicle 2200kg, initial velocity 25m/s braked for 2s shown 8s after braking

The image shows that the leading edge has more wear depth than the trailing edge when the disc is rotated in the clockwise direction. The trailing edge is represented as blue region and leading edge as red region.

When compared with Alumina A479M, the wear rate of Carbon fibre composite is comparatively less for the same conditions.

Case 2:

When the braking time of the vehicle is increased from 2 seconds to 4 seconds, the corresponding wear depth is shown in the figure:

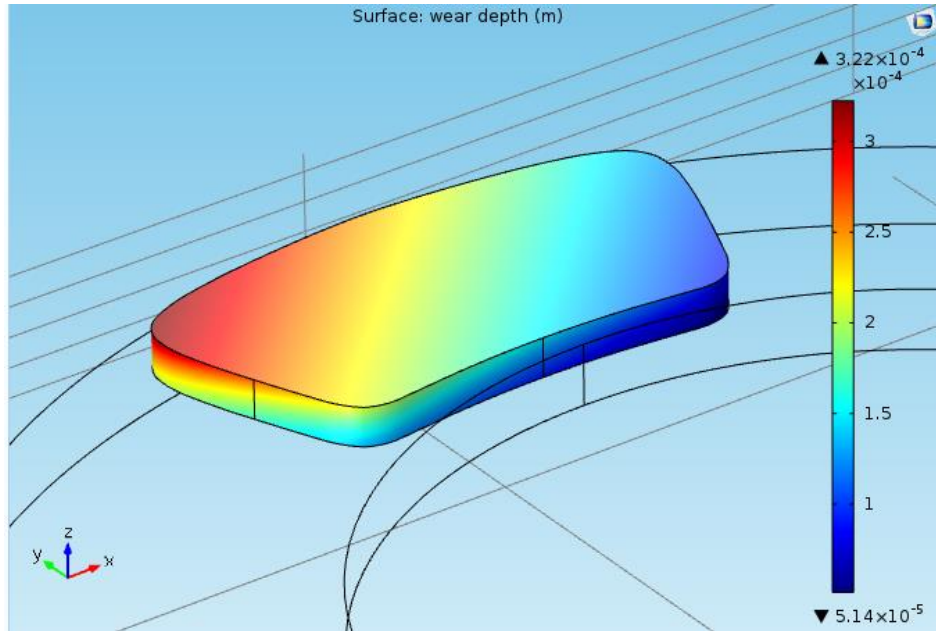


Fig 6.8: Wear depth when the braking time is increased to 4 seconds

Comparing with the earlier figure, it is clear that when the brake is applied for a longer duration the brake pad wears more and hence gives way to comparatively increased wear rate.

When compared with Alumina A479M, the wear rate of Carbon fibre composite is comparatively less for the same conditions.

Case 3:

In order to understand the impact of velocity on the brake pads we reduced the velocity by half (from 25 m/s to 15 m/s) while keeping all the other parameters same as the initially considered parameter, the corresponding wear depth is obtained.

From the results shown below, we were able to understand the impact of initial velocity on the non-uniform wear rate. We can see in the image that the uneven wear imparted on the pad is drastically reduced as a result of the reduction in velocity.

When comparing with Alumina A479M, the wear rate of Carbon fibre composite is comparatively less for the same conditions.

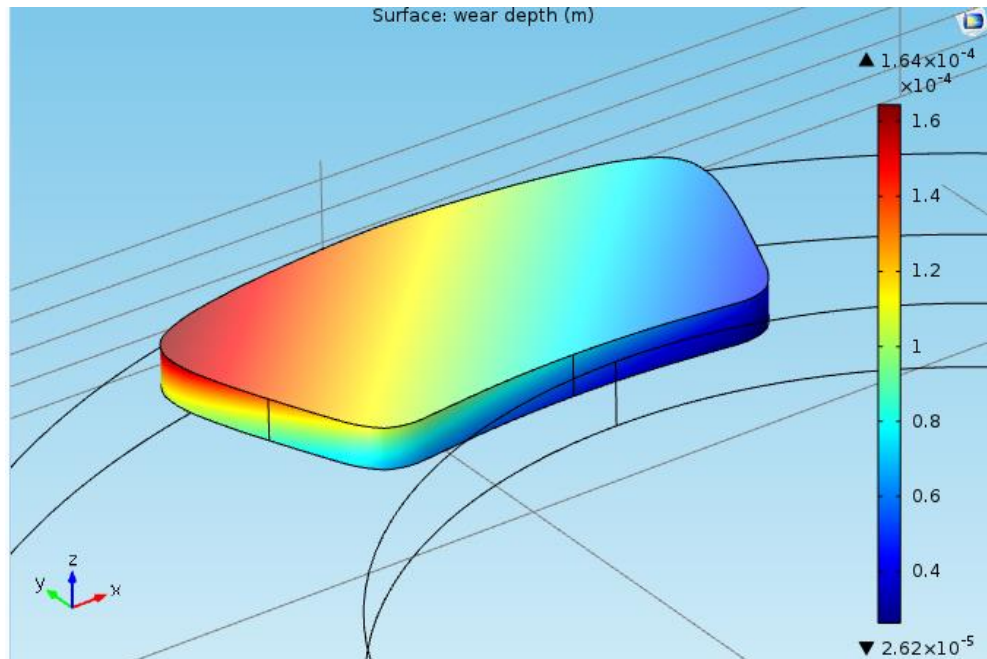


Fig 6.9: Carbon fibre composite when only the initial velocity is reduced to 15m/s

Case 4:

To understand the effect of vehicles mass on the wear depth (loaded and unloaded condition), vehicles mass is reduced to 1600 kg from 2200 kg while keeping the initial parameter same (velocity 25 m/s, Temperature 310 K)

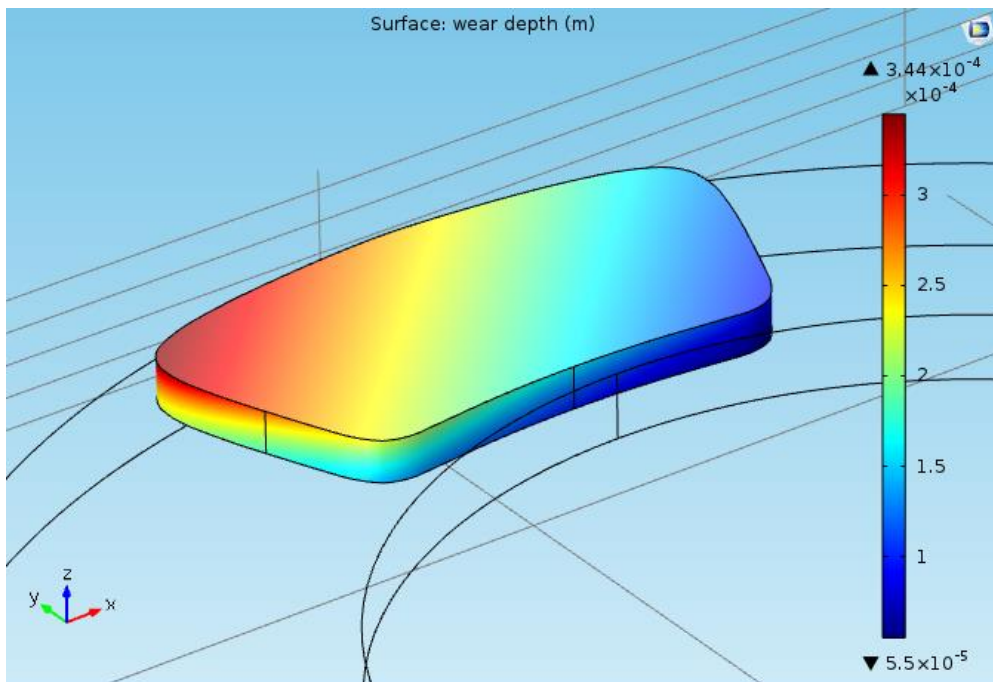


Fig 6.10: Carbon Fibre composite when only the mass is reduced from 2400kg to 1600kg

From the result above, we were able to understand the impact of mass of the vehicle on the non-uniform wear rate. From the image, we can see that though there is a reduction in the non-uniform wear rate when the mass of vehicle is reduced, we can also see that the reduction is not as drastic as the one where the initial velocity is reduced.

When comparing with Alumina A479M, the wear rate of Carbon fibre composite is very much less for the same conditions.

Case 5:

In the next case, the surrounding air temperature was reduced from 310K to 275K. This is to learn the impact of the surrounding air temperature for this particular material. The results for this particular case are shown below.

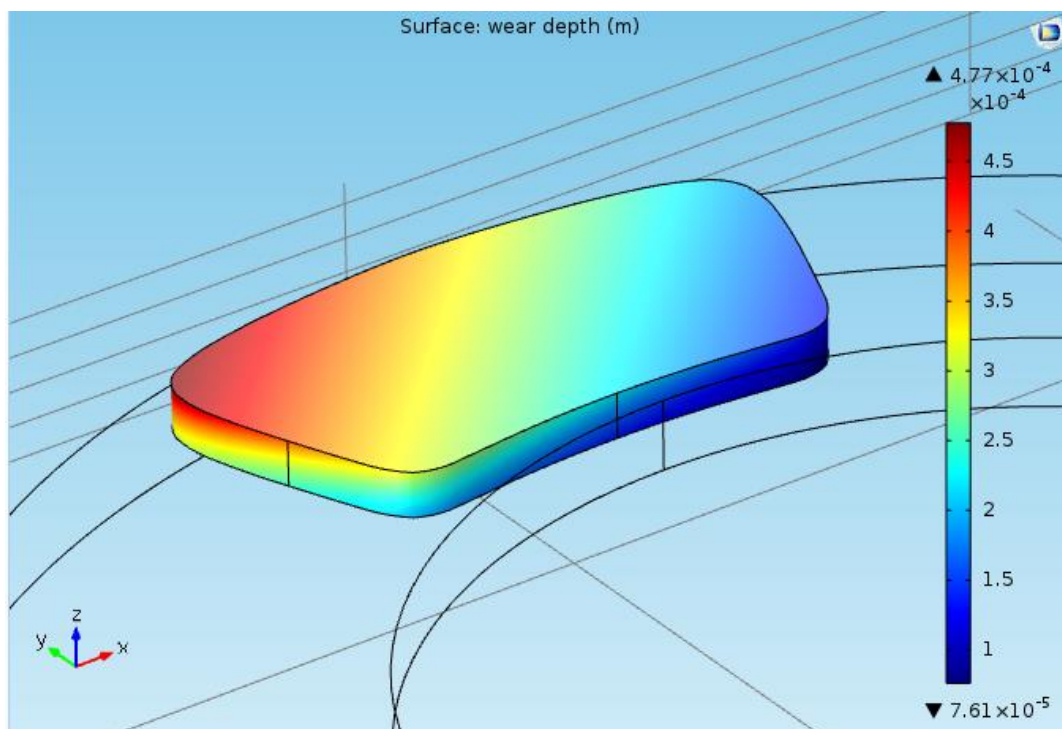


Fig 6.11: Carbon Fibre composite when the air temperature alone is reduced from 310K to 275K

The results shown above clearly indicate that the surrounding air temperature does not have any major say in the non-uniform wear rate of brake pads. The image exactly resembles the initial Fig 6.7 and so we can conclude that the surrounding air temperature has no impact on the non-uniform wear rate.

When comparing with Alumina A479M, the wear rate of Carbon fibre composite is very much less for the same conditions with a very big difference in the wear rate of these materials.

Case 6:

The image below shows the wear rate when the air temperature is 275K, mass of the vehicle is 1600kg and the initial velocity of the vehicle is 25m/s. The image also makes it apparent that the wear rate is extremely low such that it can be assumed that no significant wear takes place.

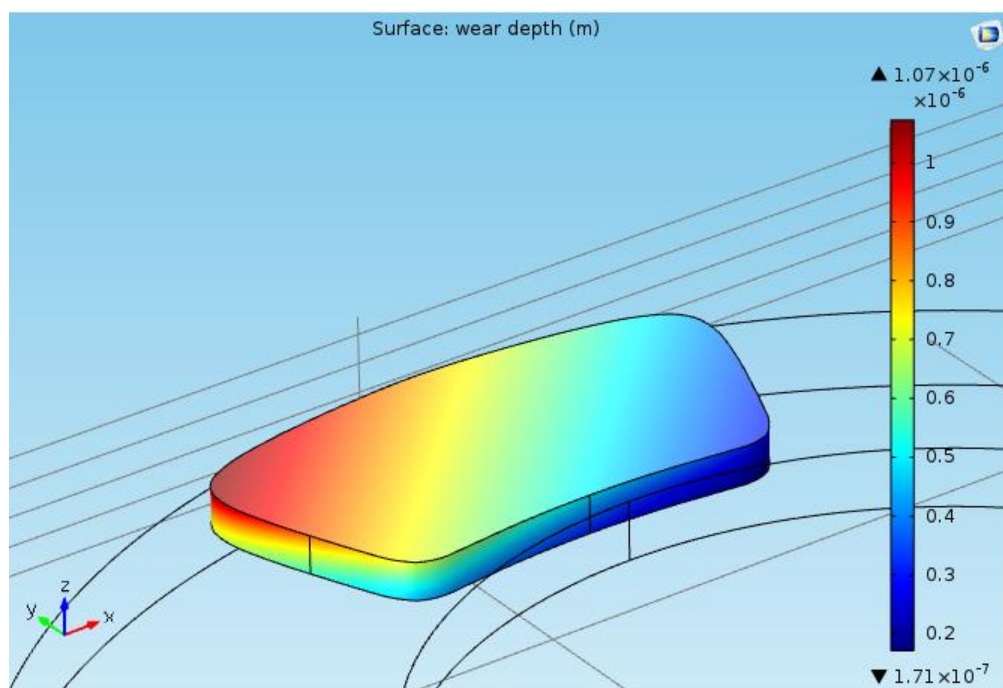


Fig 6.12: Carbon Fibre composite when the air temperature is 275K, mass of the vehicle is 1600kg and the initial velocity of the vehicle is 5m/s (condition where the wear rate is the lowest)

When comparing with Alumina A479M, the wear rate of Carbon fibre composite is very much less for the same conditions with a big difference in the wear rate of these materials.

6.3 Epoxy Resin Polyester Composite:

Case1:

The following parameters were taken into consideration, Air temperature is taken to be 310 K, Initial velocity 25 m/s, mass of the vehicle 2200 kg and the braking time as 2 sec. After

running the simulation, the following image was obtained showcasing the wear depth in the brake pad.

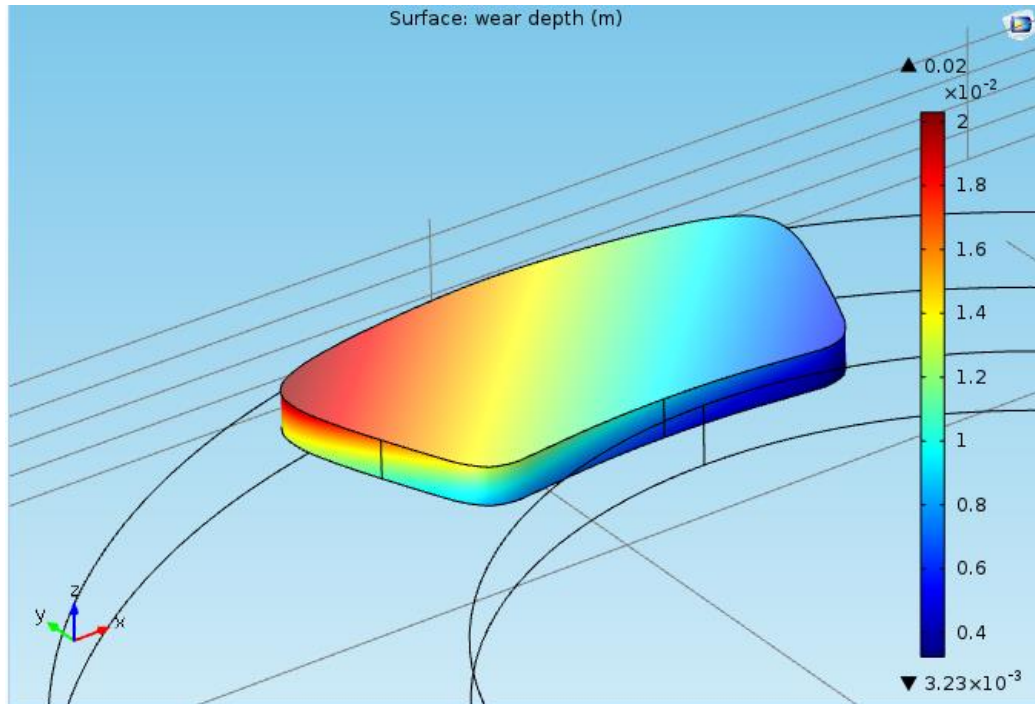


Fig 6.13: Epoxy resin polyester composite at a temperature of 310K, mass of vehicle 2200 kg, initial velocity 25m/s braked for 2s shown 8s after braking

The image shows that the leading edge has more wear depth than the trailing edge when the disc is rotated in the clockwise direction. The trailing edge is represented as blue region and leading edge as red region. Maximum wear is obtained under this condition.

When comparing with Alumina A479M and Carbon fibre composite, Epoxy resin is very much less for the same conditions with a very big difference in the wear rate of these materials.

Case 2:

When the braking time of the vehicle is increased from 2 seconds to 4 seconds, the corresponding wear depth is shown in the figure.

Comparing with the earlier figure, it is clear that when the brake is applied for a longer duration the brake pad wears more and hence gives way to comparatively increased wear rate.

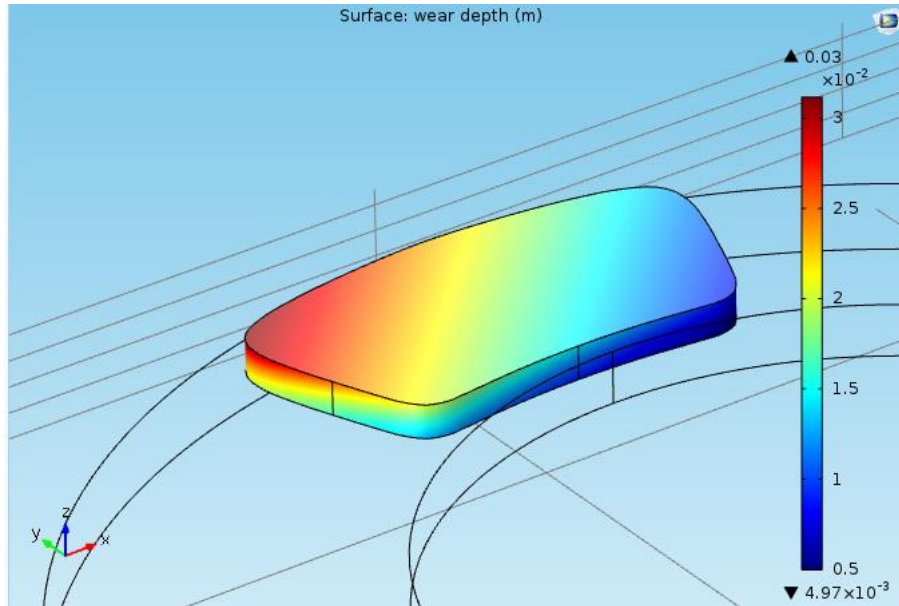


Fig 6.14: Wear depth when the braking time is increased to 4 seconds

When compared with Alumina A479M, the wear rate of Carbon fibre composite is comparatively less for the same conditions.

Case 3:

In order to understand the impact of velocity on the brake pads we reduced the velocity by half (from 25 m/s to 15 m/s) while keeping all the other parameters same as the initially considered parameter, the corresponding wear depth is obtained.

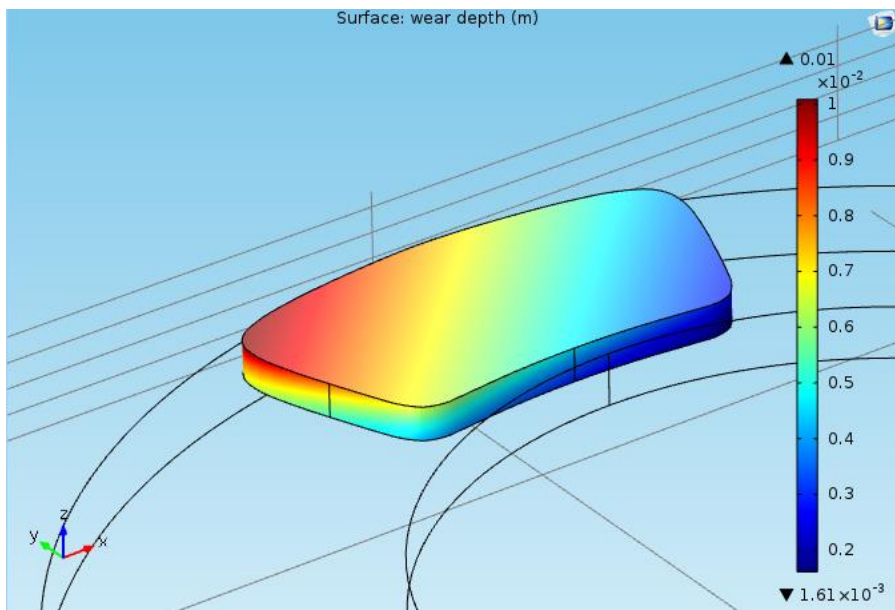


Fig 6.15: Epoxy resin polyester composite when only the initial velocity is reduced to 15m/s

From the results, we were able to understand the impact of initial velocity on the non-uniform wear rate. We can see in the image that the uneven wear imparted on the pad is drastically reduced as a result of the reduction in velocity.

When comparing with Alumina A479M and Carbon fibre composite, Epoxy resin is very much less for the same conditions with a very big difference in the wear rate of these materials.

Case 4:

To understand the effect of vehicles mass on the wear depth (loaded and unloaded condition), vehicles mass is reduced to 1600 kg from 2200 kg while keeping the initial parameter same (velocity 25 m/s, Temperature 310 K).

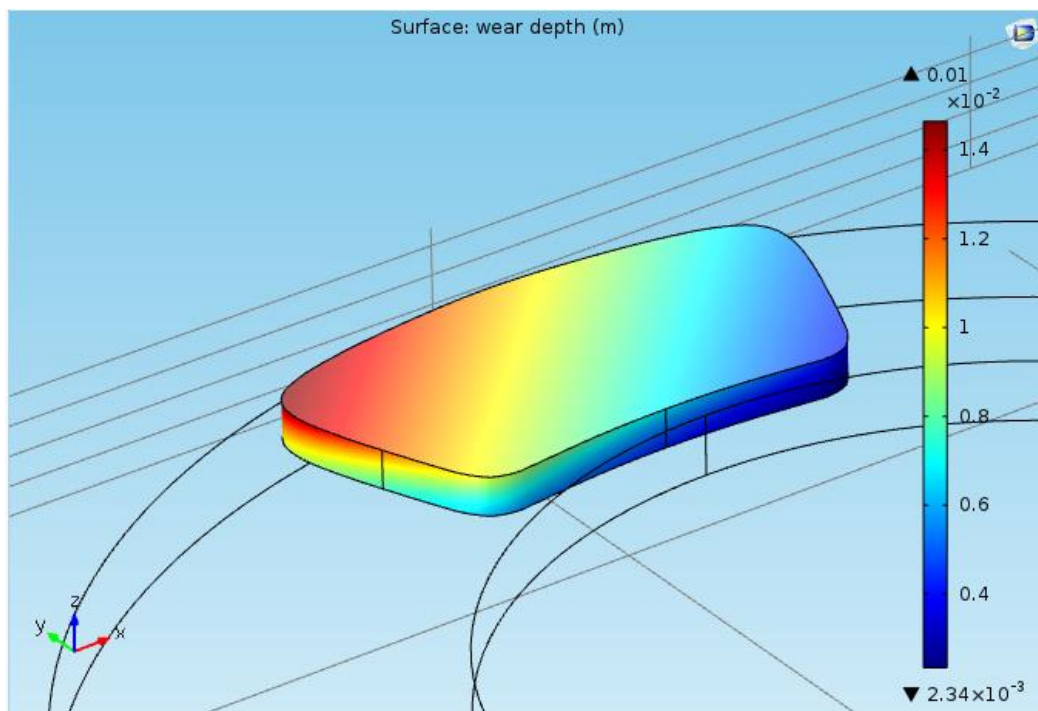


Fig 6.16: Epoxy resin polyester composite when only the mass is reduced from 2400kg to 1600kg

From the result above, we were able to understand the impact of mass of the vehicle on the non-uniform wear rate. From the image, we can see that though there is a reduction in the non-uniform wear rate when the mass of vehicle is reduced, we can also see that the reduction is not as drastic as the one where the initial velocity is reduced.

When comparing with Alumina A479M and Carbon fibre composite, Epoxy resin wear resistance is very much less for the same conditions with a very big difference in the wear rate of these materials.

Case 5:

In the next case, the surrounding air temperature was reduced from 310K to 275K. This is to learn the impact of the surrounding air temperature for this particular material. The results for this particular case are shown below.

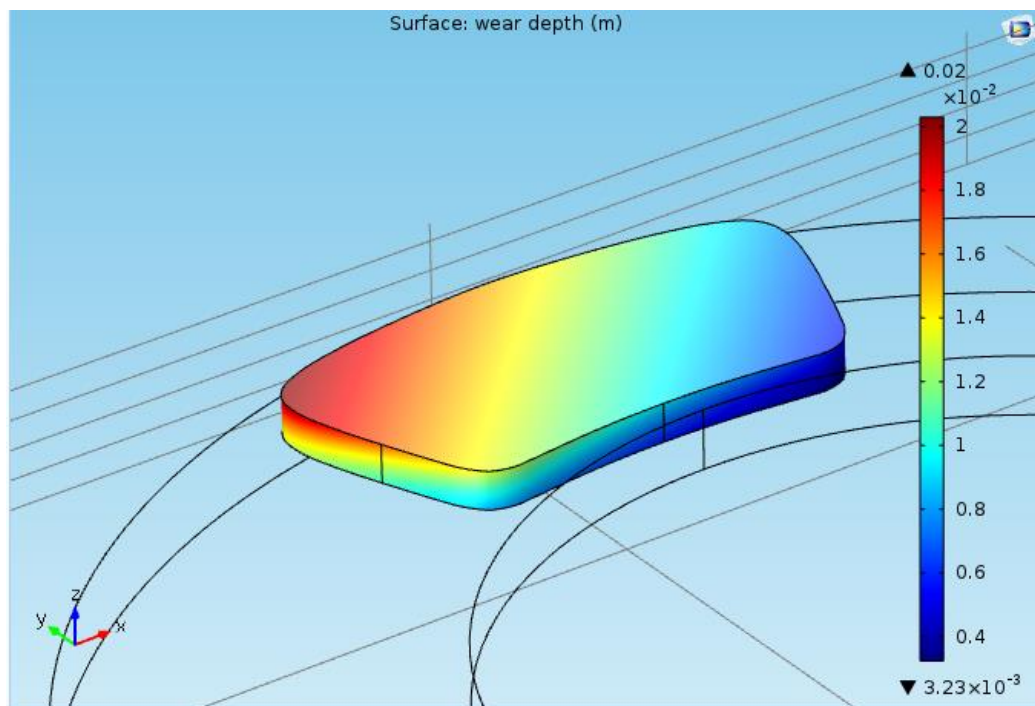


Fig 6.17: Epoxy resin polyester composite when the air temperature alone is reduced from 310K to 275

The result shown above clearly indicates that the surrounding air temperature does not have any major say in the non-uniform wear rate of brake pads. The image exactly resembles the initial Fig 6.13 and so we can conclude that the surrounding air temperature has no impact on the non-uniform wear rate.

When comparing with Alumina A479M and Carbon fibre composite, Epoxy resin wear resistance is very much less for the same conditions with a very big difference in the wear rate of these materials.

Case 6:

The image below shows the wear rate when the air temperature is 275K, mass of the vehicle is 1600kg and the initial velocity of the vehicle is 25m/s. The image also makes it apparent that the wear rate is extremely low such that it can be assumed that no significant wear takes place.

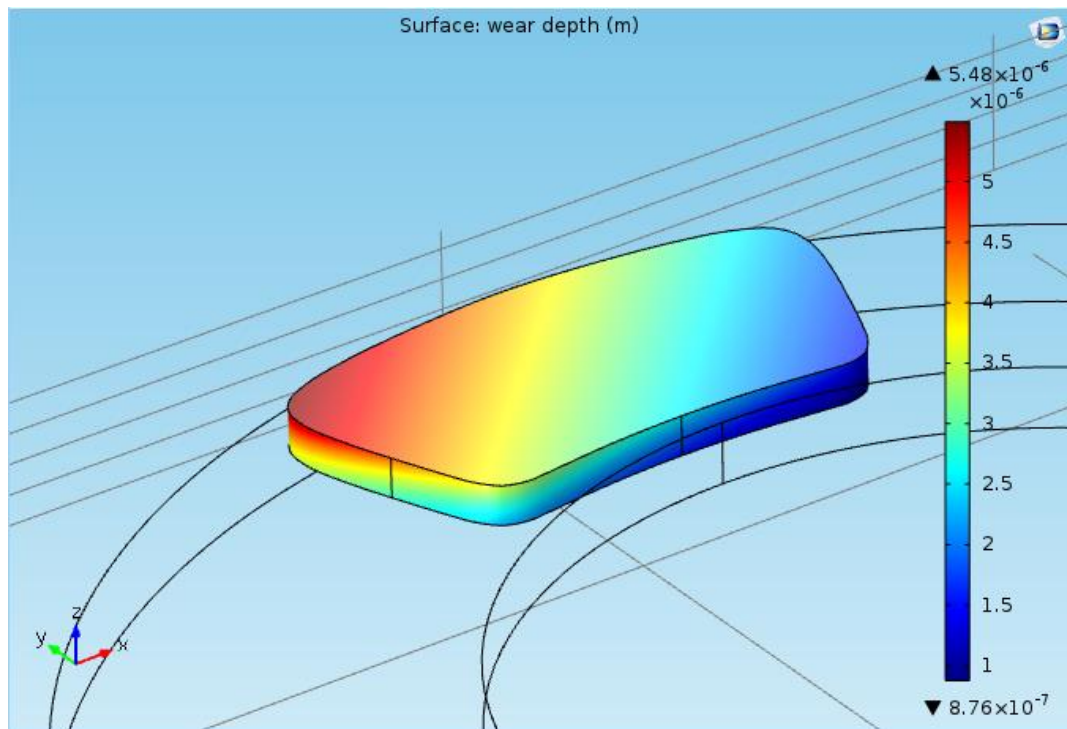


Fig 6.18: Epoxy resin polyester composite when the air temperature is 275K, mass of the vehicle is 1600kg and the initial velocity of the vehicle is 5m/s (condition where the wear rate is the lowest)

When comparing with Alumina A479M and Carbon fibre composite, Epoxy resin is very much less for the same conditions with a very big difference in the wear rate of these materials.

CHAPTER 7

CONCLUSION

- The surrounding air temperature does not have any impact on wear rate in all the cases. In case 4 in all the materials, it was found that when the temperature was varied from 310K to 275K the non-uniform wear rate in all brake pads do not change.
- For the cases where least wear occurs, epoxy resin polyester composite has the least wear resistance after alumina and carbon fibre composites.
- It is also observed that when the braking time was increased to 4 seconds, carbon fibre composite was found to wear the least followed by alumina and epoxy resin polyester composite.
- Velocity of the vehicle is found to have a greater impact on the wear rate than the mass of vehicle in all these materials.
- The rate at which the leading edge wears more than the trailing edge is the greatest for epoxy resin. So it has the highest uneven surface wearing.
- Under optimum running condition, it is evident that Carbon fibre composite wears the least when compared with the other materials and among these three; it is the most favourable brake pad material.

SCOPE OF FUTURE WORK

- Scope for calculating for larger values of wear depth.
- Scope for directly calculating the pad life span by using the wear data (by considering the braking condition).

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BIODATA

Name	Manu P. Nair
Address	Manumadhavam, House No 75M, Shanthi Garden 3 rd street, Kamaleshwaram
Date of Birth	16/10/1994
Phone Number	+919995924622
Educational details 10th: 12th: B.Tech:	Arya Central School, Pattom Arya Central School, Pattom Sree Chitra Thirunal College of Engineering, Pappanamcode
Placement Details	Nil

Name	R.Ananthakrishnan
Address	Flat No. 3-A, Pearl Gardens, Plamoodu, Pattom P.O -695004
Date of Birth	15/07/1994
Phone No.	+919746293059
Educational details 10 th : 12 th : B.Tech:	Saraswathi Vidyalaya, Vattiyoorkavu Saraswathi Vidyalaya, Vattiyoorkavu Sree chitra Thirunal College of Engineering, Pappanamcode
Placement Details	Nil

Name	Sidharth S
Address	Bhamanalam, House number-20, M P Appan Nagar, Vazhuthacaud-695014
Date of Birth	13/05/1994
Phone Number	+918891363009
Educational details 10 th : 12 th : B.Tech:	Saraswathi Vidyalaya, Vattiyoorkavu Saraswathi Vidyalaya, Vattiyoorkavu Sree Chitra Thirunal College of Engineering, Pappanamcode
Placement Details	Nil

Name	Vaishakh G
Address	Punartham, T.C-9/289-1, JNWA NB 6, Jawahar Nagar, Trivandrum-10
Date of Birth	22/11/1994
Phone Number	+919809017076
Educational details 10th: 12th: B.Tech:	Christ Nagar International School, Kowdiar Christ Nagar International School, Kowdiar Sree Chitra Thirunal College of Engineering, Pappanamcode
Placement Details	Nil